3. ISLR 4.8.3

Setting: K classes, $p = 1$, and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$ with prior π_k. The variances are **not** assumed equal.

Normal density (4.16):

$$f_k(x) = \frac{1}{\sqrt{2\pi}\,\sigma_k} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)$$

Bayes' theorem gives the posterior:

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l\, f_l(x)}$$

The denominator does not depend on k, so maximizing $p_k(x)$ is equivalent to maximizing $\pi_k f_k(x)$, and taking logs preserves the maximizer:

$$\log\big(\pi_k f_k(x)\big) = \log\pi_k - \log\sqrt{2\pi} - \log\sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

Dropping $\log\sqrt{2\pi}$, which is the same for every class, leaves the discriminant:

$$\delta_k(x) = \log\pi_k - \log\sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

Expanding $(x-\mu_k)^2 = x^2 - 2\mu_k x + \mu_k^2$:

$$\delta_k(x) = -\frac{1}{2\sigma_k^2}x^2 + \frac{\mu_k}{\sigma_k^2}x - \frac{\mu_k^2}{2\sigma_k^2} + \log\pi_k - \log\sigma_k$$

which is quadratic in x:

$$\delta_k(x) = a_k x^2 + b_k x + c_k$$

$$a_k = -\frac{1}{2\sigma_k^2}, \qquad b_k = \frac{\mu_k}{\sigma_k^2}, \qquad c_k = -\frac{\mu_k^2}{2\sigma_k^2} + \log\pi_k - \log\sigma_k$$

The boundary between classes k and m is the set where $\delta_k(x) = \delta_m(x)$:

$$(a_k - a_m)x^2 + (b_k - b_m)x + (c_k - c_m) = 0$$

The leading coefficient is

$$a_k - a_m = \frac{1}{2\sigma_m^2} - \frac{1}{2\sigma_k^2} \neq 0 \quad \text{whenever} \quad \sigma_k^2 \neq \sigma_m^2$$

so the boundary solves a genuine quadratic equation in x and is therefore not linear. The Bayes classifier is quadratic.

LDA is the special case $\sigma_1^2 = \cdots = \sigma_K^2 = \sigma^2$, which gives $a_k = -\frac{1}{2\sigma^2}$ for every k, so $a_k - a_m = 0$, the x^2 terms cancel, and the boundary becomes linear.

4. ISLR 4.8.7

Given: $\mu_{\text{Yes}} = 10$, $\mu_{\text{No}} = 0$, $\sigma^2 = 36$, $\pi_{\text{Yes}} = 0.80$, $\pi_{\text{No}} = 0.20$, and $x = 4$.

Bayes' theorem:

$$P(Y = \text{Yes} \mid X = 4) = \frac{\pi_{\text{Yes}} f_{\text{Yes}}(4)}{\pi_{\text{Yes}} f_{\text{Yes}}(4) + \pi_{\text{No}} f_{\text{No}}(4)}$$

Normal density:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Both densities share the factor $\frac{1}{\sqrt{2\pi \cdot 36}}$, which cancels:

$$P(Y = \text{Yes} \mid X = 4) = \frac{0.80 \exp\left(-\frac{(4-10)^2}{72}\right)}{0.80 \exp\left(-\frac{(4-10)^2}{72}\right) + 0.20 \exp\left(-\frac{(4-0)^2}{72}\right)}$$

$$= \frac{0.80\, e^{-0.5}}{0.80\, e^{-0.5} + 0.20\, e^{-2/9}}$$

$$= \frac{0.80(0.606531)}{0.80(0.606531) + 0.20(0.800737)} = \frac{0.485225}{0.645372}$$

$$P(Y = \text{Yes} \mid X = 4) = 0.7519$$

The probability is 0.752, or 75.2%. Since this exceeds 0.5, the Bayes classifier predicts that the company will issue a dividend.